

WORK BREAKDOWN STRUCTURE (WBS)

Public Bus Live Tracking & Crowding Estimator

NAME: SHWETHA S

SRN:PES1UG24CS452

SECTION: 5TH 'H'

1. Requirements Engineering

- Identify stakeholders and actors.
- Define the problem statement.
- Identify functional requirements.
- Identify non-functional requirements.
- Prepare the requirements traceability information.

2. Architectural Design

- Study suitable architectural styles.
- Compare Layered, Client-Server, and Microservices architectures.
- Select Microservices Architecture.
- Identify major system components.
- Define interfaces and interactions.
- Create the UML Component Diagram.
- Document the architectural justification.

3. System Design

- Design the GPS Data Injection Service.
- Design the Bus Tracking Service.
- Design the ETA Estimation Service.
- Design the Crowding Estimation Service.
- Design the Commuter Information Service.

- Design the Fleet Control Dashboard.
- Define data storage and communication between services.

4. Implementation

- Implement GPS data processing.
- Implement bus tracking functionality.
- Implement ETA calculation.
- Implement crowding estimation.
- Implement commuter information functionality.
- Implement fleet monitoring functionality.

5. Testing

- Prepare test cases.
- Test individual system functions.
- Test interactions between services.
- Test GPS and sensor data processing.
- Test ETA and crowding estimation.
- Test security and system performance.
- Fix identified defects and perform regression testing.

6. Project Management

- Create and organize the project repository.
- Create Jira project and manage work items.
- Organize tasks into sprints.
- Track bugs and project progress.
- Maintain project documentation.

7. Documentation

- Prepare Requirements Engineering documentation.
- Prepare SRS documentation.

- Prepare architectural documentation.
- Maintain Work Breakdown documentation.
- Maintain testing documentation.
- Organize all project deliverables in the GitHub repository.

Final Deliverable

A functional Public Bus Live Tracking & Crowding Estimator system with documented requirements, architecture, implementation, testing, and project management activities.